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Calculation (Euler's method for systems)

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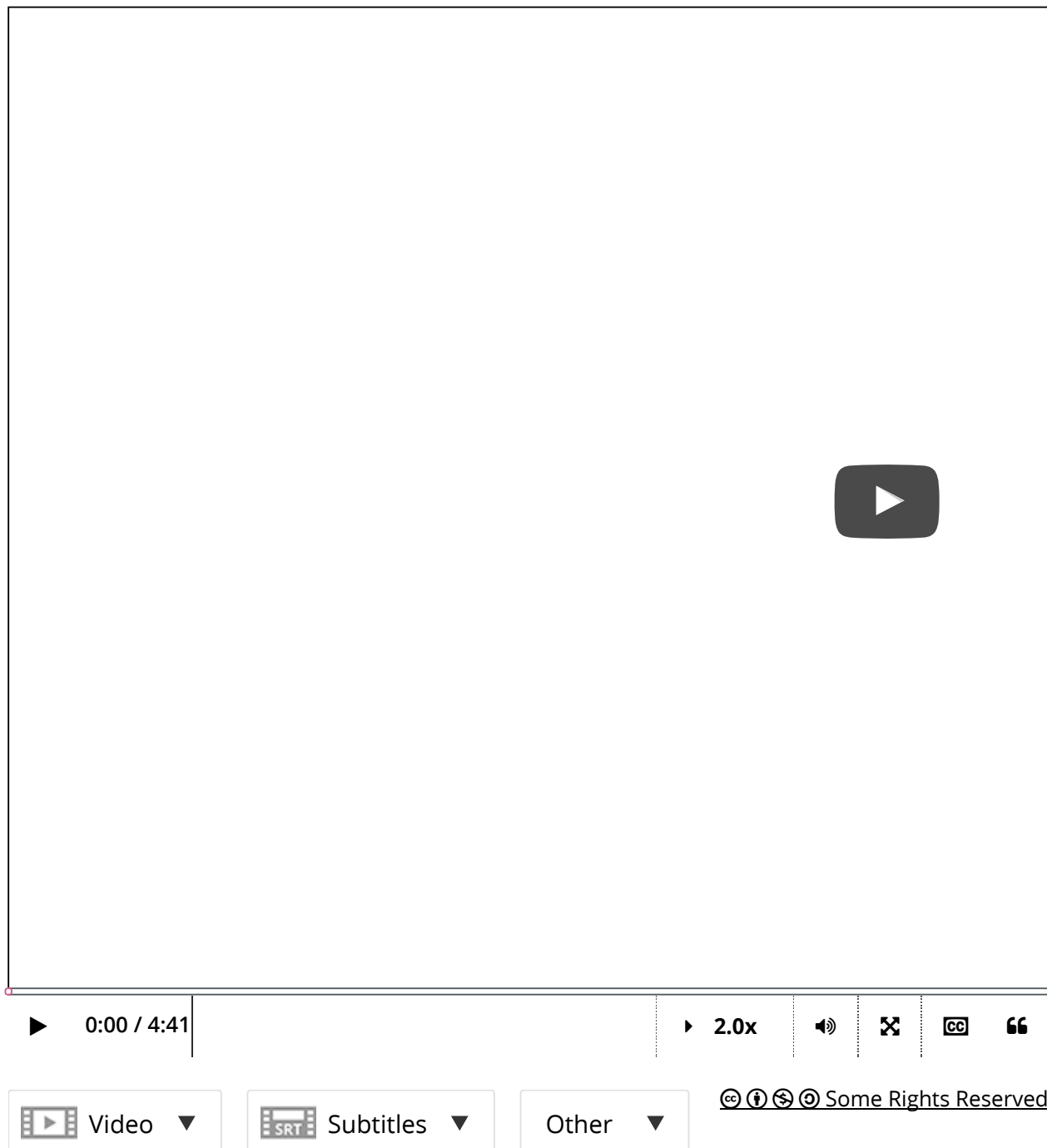
The client wants rainbowfish and gouramis in the same aquarium. That might not be a good idea, because the gouramis eat rainbowfish. Will it lead to a stable mix of the two species, or will one of them die out?

On the previous page you have seen the construction of the initial value problem for the rainbowfish population $P(t)$ and the gourami population $G(t)$:

$$\begin{cases} \frac{dP}{dt} = 0.7 P(t) - 0.007 P^2(t) - 0.04 P(t)G(t), & P(0) = 20, \\ \frac{dG}{dt} = -0.25 G(t) + 0.008 P(t)G(t), & G(0) = 5. \end{cases}$$

In the next video you will learn how you can use Euler's Method for a system of differential equations.

Euler's Method for Systems



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In the video you have learned about Euler's Method for systems of differential equations.

For the general differential equation

$$\frac{d\vec{X}}{dt} = \vec{F}(t, \vec{X}),$$

the n -th step of Euler's method is given by

$$\vec{X}((n+1)\Delta t) = \vec{X}(n\Delta t) + \Delta t \vec{F}(n\Delta t, \vec{X}(n\Delta t)),$$

in which Δt is some time step you have to choose.

Of course, doing calculations by hand quickly becomes laborious. So again we turn to a computer program. The following file contains an example code in Python for a system of differential equations:

Basic_Euler_program_systems.py

First some background information on arrays in Python.

Arrays in Python

For systems of differential equations, it is convenient to combine the different dependent variables in a vector. Here we use $\vec{X}(t) = \begin{bmatrix} P(t) \\ G(t) \end{bmatrix}$. Especially when there are many dependent variables, a vector notation makes the system often clearer and neater, both in analytical calculations and in numerical simulations.

In Python, you could try to implement vectors as lists. However, lists are not appropriate to calculate with. For example, when you try: $(3.0, 3.0)/2.0$, you will get an error message. The numpy package however, does have *arrays* with which you can implement vectors and matrices: `np.array([3.0, 3.0])/2` returns `array([1.5, 1.5])`.

Python's default mode to calculate with vectors and matrices is by element, so beware. For example, matrix $A = \begin{bmatrix} 2 & 3 \\ 4 & 5 \end{bmatrix}$ can be implemented as `A = np.array([[2, 3], [4, 5]])`.

When you calculate $A*A$, or equivalently `A**2`, Python returns `array([[ 4, 9], [16, 25]])`, so all four elements have been squared, but this is not the product of the two matrices. To calculate the product of two matrices, you need a special command: `np.dot(A,A)` which returns the correct `array([[ 16, 21], [28, 37]])`.

To do the following exercises you need to adapt the basic Euler code given above (or write your own computer program).

Exercise A

1.0/1.0 point (graded)

Use $\Delta t = 1$ and approximate with Euler's method the number of rainbowfish and gourami after ten days.

Give your answers with two decimals.

With $\Delta t = 1$, the value of $P(10) \approx$



With $\Delta t = 1$, the value of $G(10) \approx$



You have used 1 of 5 attempts

Exercise B

1/1 point (graded)

Simulate $P(t)$ and $G(t)$ for t between 0 and 100 days with $\Delta t = 1$ day.

At what time does $P(t)$ probably has its global maximum?

Give the answer in whole days.



You have used 1 of 3 attempts

Exercise C

1/1 point (graded)

We want the time step Δt at such a value that the solution P is accurate enough at the time of the probable global maximum. Use the error estimation from Section 2.3 Calculation (accuracy) to investigate which time steps give an error less than 1 rainbowfish: $y(t) - w_{\Delta t} \approx w_{\Delta t} - w_{2\Delta t}$.

Use time step $\Delta t = \frac{1}{2}$ or $\frac{1}{4}$ or $\frac{1}{8}$ or $\frac{1}{16}$ or

The largest Δt in that sequence such that the (absolute value of the) error is less than 1 rainbowfish is:

Give your answer as a fraction.



You have used 1 of 3 attempts

Exercise D

2/2 points (graded)

Using the value of Δt you found above, you can find the value of $G(10)$ and the accuracy of this value. Give your answers with two decimals.

The value of $G(10) \approx$



An estimation for the error in $G(10)$ is



You have used 1 of 5 attempts

Exercise E (1 point possible)

Question

Is the time step you have found above a good choice for both differential equations?



Answer



Reflection



Results

Correct Answer

Yes, because...

The step size you found causes each of the errors to be below 1 fish. This means the time step is small enough to attain the accuracy we want.

Your initial answer: Yes, because... (Option 1)

 "For both the differential equations the absolute error at global maximum is less than 1"

Your final answer: Yes, because... (Option 1)

 "For both the differential equations the absolute error at global maximum is less than 1"

See how your classmates answered below

Class Breakdown

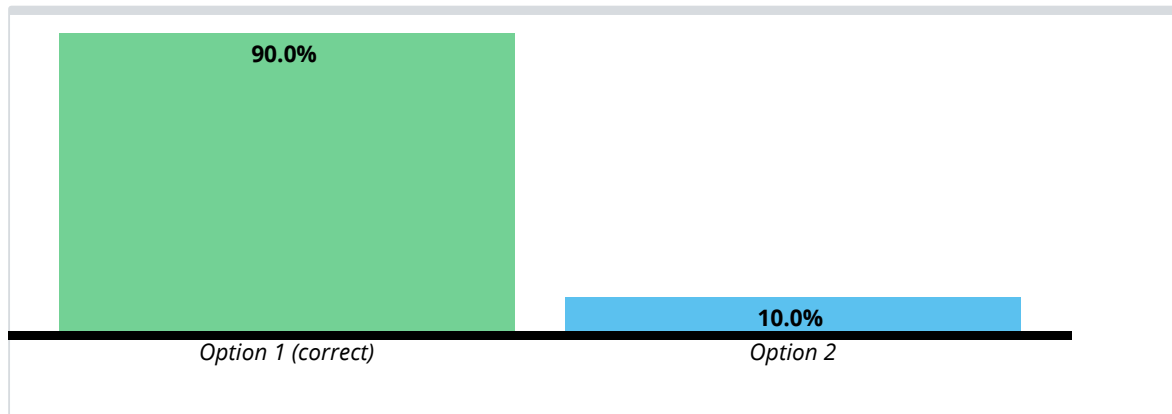
This is a look at how your classmates answered the question during the initial and final rounds.

Answer Options

Option 1 (correct)
Yes, because...

Option 2
No, because...

Initial Answer Selection



Final Answer Selection

Not enough data to generate the chart. Please check back later.

On the next page you will see how the solutions can be presented in a different way. You will plot them in the phase plane.

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English ▼

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